Angles in Polygons

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS polygon *(tapamaha)* · regular polygon *(tapamaha rite)* · interior angle · exterior angle

A The sum of the interior angles ●○○

Remember: A polygon with n sides can be cut into $n - 2$ triangles, so its interior angles add to $180 \times (n - 2)$ degrees.

1. Complete the table.

Polygon	Number of sides	Number of triangles	Sum of the interior angles
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5		
Hexagon	6		
Octagon	8		

2. Find the sum of the interior angles of a polygon with:

a) 7 sides _____ **b)** 9 sides _____ **c)** 12 sides _____

d) 20 sides _____ **e)** 15 sides _____ **f)** 102 sides _____

B Finding a missing angle ●●○

3. Work out the sum of the angles first.

a) The angles of a quadrilateral are 90°, 85°, 110° and x. Find x.

b) The angles of a pentagon are 100°, 120°, 95°, 110° and x. Find x.

c) The angles of a hexagon are 130°, 110°, 125°, 140°, 95° and x. Find x.

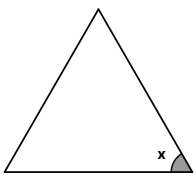
d) The angles of a pentagon are 108°, 108°, 108°, 108° and x. Find x.

C Regular polygons ●●○

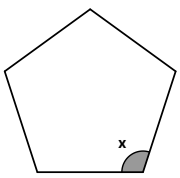
Remember: In a regular polygon all the sides are equal and all the angles are equal. So each interior angle is the angle sum divided by the number of sides.

4. Each shape is a regular polygon. Find the size of the angle marked x.

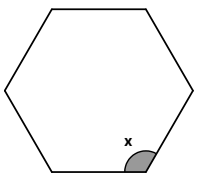
a)



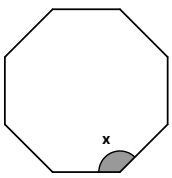
b)



c)



d)



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5. Find the size of each interior angle of:

a) A regular polygon with 9 sides

b) A regular polygon with 10 sides

c) A regular polygon with 12 sides

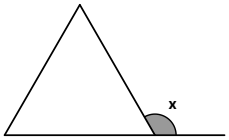
d) A regular polygon with 20 sides

D Exterior angles ●●○

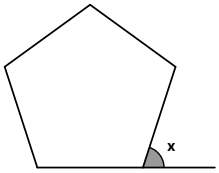
Remember: The exterior angles of any polygon add to 360°. At each corner, the interior angle and the exterior angle add to 180°.

6. Each shape is a regular polygon with one side made longer. Find the size of the exterior angle marked x.

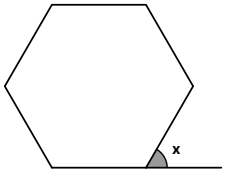
a)



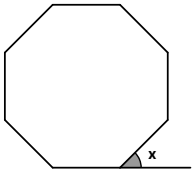
b)



c)



d)



7. Find the size of each exterior angle of:

a) A regular polygon with 9 sides

b) A regular polygon with 12 sides

c) A regular polygon with 20 sides

d) A regular polygon with 36 sides

8. Answer each question.

a) An interior angle of a polygon is 110°. Find the exterior angle at the same corner.

b) An exterior angle of a polygon is 45°. Find the interior angle at the same corner.

c) An interior angle of a polygon is 162°. Find the exterior angle at the same corner.

E How many sides? ●●●

Remember: The exterior angles add to 360°, so the number of sides of a regular polygon is 360 divided by one exterior angle.

9. How many sides does each regular polygon have?

a) Each exterior angle is 60°.

b) Each exterior angle is 40°.

c) Each exterior angle is 30°.

d) Each exterior angle is 72°.

e) Each interior angle is 120°.

f) Each interior angle is 135°.

g) Each interior angle is 150°.

h) Each interior angle is 108°.

★ Extension

10. Can a regular polygon have interior angles of 100°? Show your working.

11. The interior angles of a polygon add to 1,260°. How many sides does it have?

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Angles in Polygons

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS polygon *(tapamaha)* · regular polygon *(tapamaha rite)* · interior angle · exterior angle

A The sum of the interior angles ●○○

Remember: A polygon with n sides can be cut into $n - 2$ triangles, so its interior angles add to $180 \times (n - 2)$ degrees.

1. Complete the table.

Polygon	Number of sides	Number of triangles	Sum of the interior angles
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5		
Hexagon	6		
Heptagon	7		

2. Find the sum of the interior angles of a polygon with:

a) 8 sides _____ **b)** 10 sides _____ **c)** 12 sides _____

d) 20 sides _____ **e)** 18 sides _____ **f)** 52 sides _____

B Finding a missing angle ●●○

3. Work out the sum of the angles first.

a) The angles of a quadrilateral are 90°, 95°, 105° and x. Find x.

b) The angles of a pentagon are 105°, 115°, 100°, 120° and x. Find x.

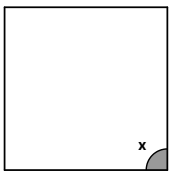
c) The angles of a hexagon are 120°, 115°, 130°, 135°, 100° and x. Find x.

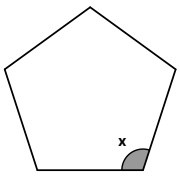
d) The angles of a hexagon are 120°, 120°, 120°, 120°, 120° and x. Find x.

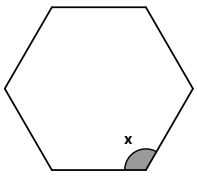
C Regular polygons ●●○

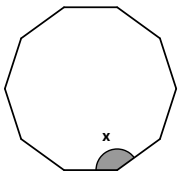
Remember: In a regular polygon all the sides are equal and all the angles are equal. So each interior angle is the angle sum divided by the number of sides.

4. Each shape is a regular polygon. Find the size of the angle marked x.

a) 

b) 

c) 

d) 

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5. Find the size of each interior angle of:

a) A regular polygon with 8 sides

b) A regular polygon with 9 sides

c) A regular polygon with 12 sides

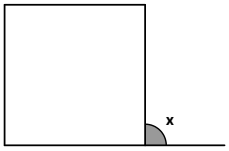
d) A regular polygon with 18 sides

D Exterior angles ●●○

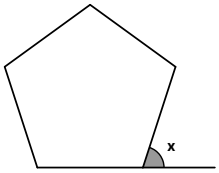
Remember: The exterior angles of any polygon add to 360°. At each corner, the interior angle and the exterior angle add to 180°.

6. Each shape is a regular polygon with one side made longer. Find the size of the exterior angle marked x.

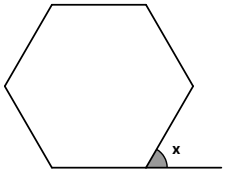
a)



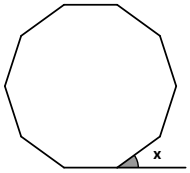
b)



c)



d)



7. Find the size of each exterior angle of:

a) A regular polygon with 10 sides

b) A regular polygon with 12 sides

c) A regular polygon with 18 sides

d) A regular polygon with 30 sides

8. Answer each question.

a) An interior angle of a polygon is 125°. Find the exterior angle at the same corner.

b) An exterior angle of a polygon is 40°. Find the interior angle at the same corner.

c) An interior angle of a polygon is 156°. Find the exterior angle at the same corner.

E How many sides? ●●●

Remember: The exterior angles add to 360°, so the number of sides of a regular polygon is 360 divided by one exterior angle.

9. How many sides does each regular polygon have?

a) Each exterior angle is 90°.

b) Each exterior angle is 45°.

c) Each exterior angle is 36°.

d) Each exterior angle is 24°.

e) Each interior angle is 120°.

f) Each interior angle is 140°.

g) Each interior angle is 144°.

h) Each interior angle is 160°.

★ Extension

10. Can a regular polygon have interior angles of 130°? Show your working.

11. The interior angles of a polygon add to 1,620°. How many sides does it have?

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